Verification and Validation Plan

Adaptive Retail Advertising System

Privacy-First Edge Adaptive Signage MVP

Document ID	VVP-ADAPTIVE-EDGE-MVP-v1.0
Purpose	Define the complete verification and validation program, close remaining gaps, and establish sales-release criteria once the system passes.
Governing Basis	Golden Build Plan v1.1, Roadmap Addendum v1, ICD set, SAD, and Technical Requirements Package.
Intended Use	Internal engineering, pilot validation, investor/customer diligence, and pre-sales readiness package.

Date: 2026-03-17

1. Executive Summary

This plan converts the existing architecture, ICDs, Golden Build Plan, Roadmap Addendum, and Technical Requirements Package into a field-ready V&V program. Its purpose is not merely to confirm that the software works in a lab; it establishes the evidence package required to prove that the product is safe to deploy, stable enough to support, commercially credible, and ready to enter a paid pilot or first-sale motion.

Sales-release principle: the system is commercially ready only when it passes technical verification, pilot validation, operator usability checks, privacy audits, and business-value proof against a static baseline using the same approved content library.

2. Governing Basis and Scope

This V&V plan inherits all locked project constraints. It does not redefine architecture. It verifies compliance against the current baseline.

- Golden Build Plan v1.1: locked product definition, privacy boundary, MVP scope, service architecture, phased deployment order, reliability law, pilot experiment requirement, and success metrics.
- Roadmap Addendum v1: locked adaptive variant policy, dashboard surface, manual approval requirement, non-bypassable approvals, and operator controls.
- Consolidated ICD + Interface Addendum: versioned interface contracts, internal/external exposure rules, metadata-only privacy posture, and supervisor/network requirements.
- Technical Requirements Package: measurable performance targets, acceptance tests, availability and reliability thresholds, calibration requirements, deployment constraints, observability requirements, and retention/security expectations.

3. V&V Objectives

Objective	Question Answered	Primary Evidence	Release Impact
Requirement verification	Did the system meet each locked technical requirement?	Traceability matrix, automated tests, measured logs	Blocks technical release
System validation	Does the appliance behave correctly in the target store environment?	Pilot scene tests, soak tests, operator walkthroughs	Blocks pilot deployment
Commercial validation	Does adaptive mode create enough value versus static rotation?	A/B pilot analysis, owner feedback, support burden review	Blocks first paid rollout
Compliance assurance	Did the system preserve privacy, approvals, and local-first boundaries?	Storage audit, egress audit, approval bypass tests, network scans	Blocks customer-facing claims

4. Test Strategy by Level

Level	Purpose	Owner	Entry	Exit
Unit	Verify local logic, schema enforcement, safety rules, approval logic, state machine transitions.	Service owner	Code complete	All critical unit tests green
Contract	Verify ICD conformance, versioning, privacy flags, ordering, idempotency, and reject behavior.	Backend + CV + frontend	Schemas frozen	Contract suite green on CI
Integration	Verify services interoperate across camera, CV, audience-state, decision, creative, player, dashboard, DB, and supervisor.	System integrator	Containers deployable	Cross-service scenarios pass
System	Verify whole-appliance behavior including boot, playback continuity, privacy, recovery, updates, and observability.	Founder/system owner	Stable integrated image	System campaign passes
Pilot validation	Verify target-store usability and business value against static baseline.	Founder + pilot customer	System campaign passed	Pilot exit criteria achieved
Commercial release	Verify supportability, documentation, install repeatability, and sales-proof package.	Founder	Pilot validation passed	Release board approves first-sale status

5. Requirement Traceability and Compliance Mapping

The matrix below ensures this plan remains compliant with the Golden Build Plan and Roadmap Addendum rather than drifting into a new architecture.

Source Requirement	Source Doc	Verification Method	Primary Evidence	Status to Release
Screen must never go blank; fallback behavior must exist.	Golden Build Plan / TRD	Boot, switch, fault-injection, player recovery, fallback creative tests	AT-BOOT, AT-SWITCH, AT-FAULT logs and video capture	Hard release gate
No raw images, clips, biometric identifiers, or embeddings stored or egressed.	Golden Build Plan / ICD / TRD / SAD	Filesystem scan, DB scan, packet capture, config inspection, negative tests	Privacy audit report	Hard release gate
Adaptive behavior must stay within approved policy space and manual approval remains mandatory by default.	Roadmap Addendum / ICD-5 / ICD-6	Approval bypass tests, manifest policy validation, dashboard workflow tests	Approval control test pack	Hard release gate
WAN-independent runtime; remote admin via WireGuard only.	Golden Build Plan / ICD-NET-1 / TRD	LAN/WAN isolation test, VPN access test, firewall scan	Network verification report	Hard release gate
Adaptive mode must be compared to static mode using same approved library.	Golden Build Plan	Pilot A/B design and analysis	Pilot business validation report	Sales gate
Owner must feel in control; manual override and safe mode must always exist.	Golden Build Plan / Roadmap Addendum	Dashboard usability walkthrough, override tests, operator SOP drill	Operator acceptance signoff	Sales gate

6. Verification Campaigns

6.1 Build-Phase Verification by Deployment Sequence

Campaign	Scope	Key Checks	Pass Criteria	Evidence	Gate Type
Phase 0 Platform Baseline	JetPack, Docker, WireGuard, Chromium, boot services	Boot to managed runtime, remote admin only via VPN, no public exposure	Boot-to-runtime verified; VPN works; WAN not required for playback shell	Provisioning checklist + scan report	Build gate

Campaign	Scope	Key Checks	Pass Criteria	Evidence	Gate Type
Phase 1 Vision Stack	input-cv + DeepStream + tracker	Stable ingest, no image persistence, camera reconnect, metadata-only output	24h stable ingest; privacy flags false; reconnect works	CV bring-up log + storage audit	Build gate
Phase 2 Audience State	smoothing and confidence logic	Stable state transitions, unknown handling, replay consistency	Replay outputs deterministic; low-confidence suppression works	Replay report	Build gate
Phase 3 Player Runtime	React/Vite player + kiosk mode + fallback bundle	Always-on display, asset preload, fallback route, no flicker baseline	No blank on controlled switch; fallback renders offline	Player validation capture	Build gate
Phase 4 Decision Engine	rules-first decisioning	Keep/switch/ fallback logic, explainability, stale state handling	Decision logs explain every switch; stale inputs rejected	Decision trace pack	Build gate
Phase 5 Creative Assembly	manifest generation and policy enforcement	Approved-only rendering, manifest integrity, cache miss behavior	Unapproved manifest always rejected; cache miss falls back safely	Creative conformance report	Build gate
Phase 6 DB and Optimizer	event storage, optimizer, metrics	Append-only events, cold-start fallback, no DB dependency for playback	DB outage does not blank screen; events replayable	Persistence outage test	Build gate
Phase 7 Dashboard	owner control plane	Asset upload, approvals, analytics, safe mode controls	Approval workflow complete; manual override works	Dashboard UAT	Pilot entry gate
Phase 8 Reliability Hardening	supervisor, health checks, maintenance	Restart ladder, maintenance window logic, 72h run	24-72h stable; restart ladder proven	Reliability campaign report	Pilot entry gate

6.2 Core System Verification Campaigns

Campaign	Scope	Key Checks	Pass Criteria	Evidence	Gate Type
C1 Boot and startup	Cold boot through operational state	Time to first frame; time to operational; services healthy	TTFF <= 60 s; operational <= 120 s	Boot trace + timestamp log	Hard gate
C2 Playback continuity	Normal operation and planned switches	Black-screen duration; command apply timing; fallback route	0 target; never > 250 ms blank	Screen recording + event log	Hard gate
C3 CV performance	Presence, counts, latency, stability	Recall, FP rate, count MAE, count within +/-1, p95 latency	Meet TRD thresholds	Labeled-scene report	Hard gate
C4 Reliability and recovery	Service failures and degraded operation	Kill CV, decision, creative, player, DB; verify fallback behavior	Playback preserved; player recovers <= 10 s; ladder works	Fault-injection report	Hard gate
C5 Thermal/resource	Load under representative use	GPU/CPU/RAM/ storage budgets, thermal headroom, degrade/protect states	Meet TRD budgets; no throttling-induced visible failure	72h soak + tegrastats analysis	Hard gate
C6 Update and rollback	Controlled pilot release updates	Signed bundle acceptance, rollback, no-brick behavior	Unsigned rejected; rollback works; no unrecoverable state	Update drill report	Hard gate
C7 Privacy and network	Storage, egress, exposure, VPN-only admin	No visual persistence/egress; LAN/WAN exposure correct	All privacy and network tests pass	Privacy/ network audit	Hard gate

7. Detailed Test Matrix

The following matrix consolidates the most important pass/fail criteria needed for first pilot and first sale.

ID	Test	Method	Metric	Target	Severity if Fail	Release Impact
AT-BOOT	Cold boot to content	System test	TTFF / operational ready	<=60 s / <=120 s	Critical	No pilot
AT-SWITCH	Controlled switching	System test + video	Blank interval / flicker	0 target, never >250 ms	Critical	No pilot
AT-CV-LAT	Capture to metadata latency	Instrumented replay	p95 latency	<=150 ms @ 10 FPS	Critical	No pilot
AT-CV-ACC	Pilot-scene CV accuracy	Labeled replay	Recall / FP / count MAE	>=0.95 / <=1 per 10 min / <=0.6	Critical	No pilot
AT-SOAK-72H	Long run stability	72h soak	Runtime stability / resource budget	No crash requiring manual intervention	Critical	No sale
AT-FAULT	Failure ladder	Fault injection	Recovery behavior	Playback preserved; player <=10 s	Critical	No pilot
AT-PRIV-STORE	No visual retention	Audit	Raw imagery count	0	Critical	No pilot
AT-PRIV-EGRESS	No raw video egress	Pcap + firewall audit	Frame egress count	0	Critical	No pilot
AT-APPROVAL	Approval bypass resistance	API/UI abuse tests	Unauthorized runtime eligibility	0 bypass paths	Critical	No sale
AT-OVERRIDE	Operator control	UAT drill	Override to safe mode	100% successful	Major	No sale
AT-UPDATE	Update safety	Drill	Rollback success	100% on tested scenarios	Critical	No sale
AT-STATIC-VS-ADAPTIVE	Commercial value pilot	Pilot A/B	Dwell uplift / owner acceptance	Positive direction + acceptable support burden	Critical	No sale

8. Validation in the Target Pilot Environment

Validation confirms that the system is useful in the intended deployment context: a small, cooperative retail site such as a coffee shop or quick-service environment with one camera, one display, moderate traffic, and relatively stable indoor lighting.

Validation Area	Scenario	Measure	Success Threshold	Why It Matters
Scene variability	Daylight, low light, glare, occlusion, camera jitter	Stability and CV thresholds	All mandatory scenario classes pass	Prevents lab-only success
Operator usability	Setup, approval workflow, safe mode, analytics check	Task completion without founder intervention	Owner can perform core flows after brief instruction	Reduces support burden
Commercial signal	Static vs adaptive on same content library	Dwell / attention delta, owner feedback	Adaptive shows positive direction and no trust erosion	Supports sales narrative
Operational fit	Open hours, off-hours update window, network interruptions	Playback continuity and dashboard availability	No visible failure during realistic interruptions	Confirms appliance behavior
Supportability	Remote diagnosis over WireGuard	Time to triage common issues	Founder can diagnose remotely within acceptable time	Solo-founder feasibility

9. Gap Closure Items Not Fully Covered by Existing TRD

The TRD already covers technical requirements and acceptance tests. The following items expand coverage where the current document set is still thin and where commercial readiness would otherwise remain weak.

Gap	Why It Matters	Verification Activity	Evidence	Owner
Commercial value threshold not explicitly operationalized	A technically good system can still fail sales if the value proof is weak	Run a pilot A/B protocol with pre-declared metrics and founder debrief	Pilot business validation report	Founder
Support burden under real pilot conditions	Solo-founder economics can fail even if product works	Log all support tickets, time spent, on-site visits avoided, remote resolution rate	Support burden ledger	Founder

Gap	Why It Matters	Verification Activity	Evidence	Owner
Install repeatability for non-technical customers	Sales readiness requires predictable deployment	Execute timed install dry-runs using only install guide and provisioning checklist	Installation verification report	Founder
Camera SKU and mounting repeatability	Real-world CV quality depends on repeatable optics and placement	Validate at least one chosen camera and mounting SOP across multiple installs	Camera qualification report	Founder
Sales-proof evidence package	Customers need concise proof, not engineering raw logs	Assemble a customer-safe proof pack with uptime, privacy, approvals, and pilot uplift summary	Sales readiness binder	Founder

10. Commercial Readiness and Sales Release Gate

Passing engineering tests alone is not sufficient for first sale. The system becomes sales-ready only when all technical and commercial release criteria below are met.

Gate	Minimum Requirement	Evidence Required	Decision
Technical gate	All hard-gate tests pass; no unresolved critical defects	Signed V&V summary and failed-test disposition log	Pass/hold
Privacy gate	No image retention, no frame egress, no approval bypass path, VPN-only admin exposure	Privacy + network audit pack	Pass/hold
Operational gate	72h stable run, restart ladder proven, install <= planned window, remote triage workable	Reliability report + install drill + support drill	Pass/hold
Operator gate	Owner can upload, approve, enable, pause, safe-mode, and review analytics without engineering help	UAT signoff + task checklist	Pass/hold

Gate	Minimum Requirement	Evidence Required	Decision
Commercial gate	Adaptive mode shows positive value signal over static mode and customer expresses willingness to continue/pay	Pilot business validation report + customer note	Pass/hold

Recommended first-sale rule: Do not sell as an always-improving AI platform until the first pilot proves stable operation, approval-safe adaptive behavior, and a positive adaptive-versus-static signal using the same approved content library.

11. Evidence Package and Deliverables

- V&V summary report with pass/fail status by gate.
- Traceability matrix mapping every hard requirement to evidence.
- Boot, switch, fault-injection, privacy, and update drill reports.
- 72h soak report with tegrastats/resource graphs and defect log.
- Pilot validation report with static versus adaptive comparison.
- Support burden report: time spent, issues encountered, remote resolution rate.
- Customer-safe sales proof pack: privacy posture, uptime, approval controls, and pilot outcome summary.
- Install/provisioning checklist and rollback procedure.

12. Exit Criteria by Milestone

Milestone	Entry	Exit Criteria	Artifacts	Go/No-Go Owner
Integrated system complete	All major services running on target hardware	All build-phase verification campaigns complete	Build verification pack	Founder
Pilot-ready	Integrated system complete	All hard technical/privacy/network gates pass; operator UAT ready	Pilot entry memo	Founder
Pilot complete	Pilot-ready and customer site active	Static vs adaptive report, support burden reviewed, customer feedback collected	Pilot closeout report	Founder
Sales-ready	Pilot complete	All commercial gates pass and no unresolved release blockers remain	Sales release memo	Founder

13. Recommended Immediate Next Actions

- Implement the automated hard-gate suite first: boot, switch, CV latency, privacy storage/egress, fault injection, update rollback.
- Create the pilot-scene labeled replay set and lock the calibration report template.
- Write the pilot protocol for static versus adaptive comparison before any pilot starts.
- Prepare the customer-safe proof pack template now so evidence collection matches the sales story.
- Qualify one camera SKU, one mount, and one installation SOP before seeking repeat pilots.

Appendix A. Release Gate Summary

Category	Mandatory to Ship?	Release Standard
Blank screen prevention	Yes	No visible blank > 250 ms in normal operation
Privacy	Yes	0 retained imagery, 0 frame egress
Approvals	Yes	0 approval bypass paths; approved-only runtime use
Availability	Yes	>=99.5% playback availability target for pilot
Reliability	Yes	72h stable run and restart ladder proven
Operator control	Yes	Manual override and safe mode fully usable
Commercial value	Yes for first sale	Adaptive mode shows positive signal vs static with acceptable support burden